

Q # 1 Answer the following?

(marks 5*5=25)

1. Define and Elaborate agents in AI with examples (any two)?
2. What is expert systems? Elaborate your answer with two examples?
3. What is uncertainty? Briefly describe the fuzzy logic in predicting uncertainty?
4. Present a comprehensive comparative analysis of different supervised learning techniques?
5. Briefly describe the concept of pruning? why and where it is implemented in AI?

Q # 2 Solve it with the help of any unsupervised learning technique. Assume Centre points are (in case of solving thorough K-means clustering): (1,1) and (5,7) (marks 5)

subject	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Pass 1:
1 - A
2 - B
3 - A
4 - C
5 - D
6 - B
7 - C
8 - D
9 - B
10 - C
Pass 2:
1 2 3 4 5 10 9 7 6 8
Pass 3:
1 2 3 4 5 6 7 8 9 10
15, 25, 28, 30, 35, 40, 45, 50, 55
60, 70

Q # 3 Apply chaining techniques on given facts based on following rules

(marks 5)

Rules:

R1: if hot AND smoky Then ADD fire

R3: If fire then Add switch_on_sprinklers

R5: If Sprinklers_on Then DELETE dry

Facts:

F1: alarm_beeps

F2: hot

Goal: Should I switch sprinklers on?

R2: If alarm_beeps Then Add smoky

R4: If dry Then ADD switch_on_humidifier

Q # 4 Implement A* search algorithm on the following graph and find the best route starting from S to Goal G. (marks 5)

